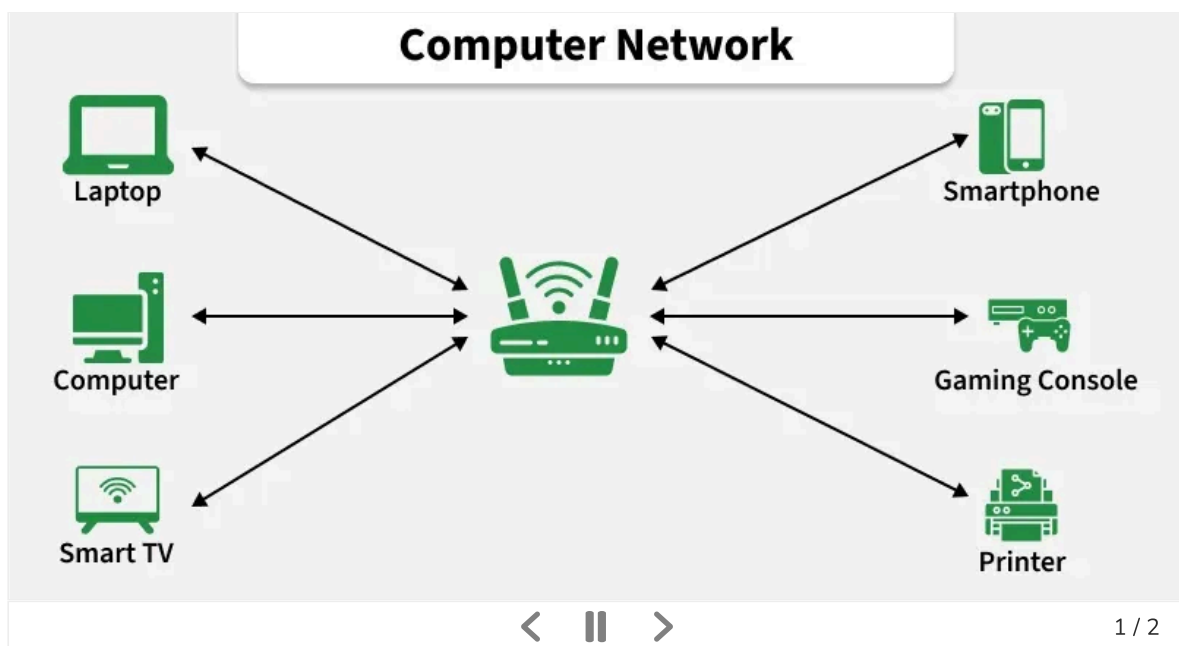


Computer Network Tutorial

Last Updated : 9 Feb, 2026

A computer network is a system of interconnected devices, such as computers, servers, smartphones, and printers, that communicate to exchange data. They communicate using wired or wireless connections and can range from small home networks to the global Internet.

- **Resource Sharing:** Allows sharing of hardware like printers, scanners, and storage devices, reducing cost.
- **Data Sharing:** Enables users to share files, applications, and databases easily.
- **Communication:** Supports email, video conferencing, instant messaging, and web access.
- **Data Management:** Helps in storing, securing, and backing up data efficiently.
- **Remote Access:** Allows users to access systems and cloud services from anywhere.
- **Better Collaboration:** Improves teamwork and information exchange among users.



Fundamentals of Computer Networks

Computer networking starts with these fundamental concepts. It explains what a computer network is, its benefits, key characteristics, and the essential hardware components involved.

- [Basics of Computer Networks](#)
- [Network Devices](#)
- [Types of Networks](#)

Network Models

Computer networks rely on these foundational frameworks to standardize communication. It explains the layered structures of the OSI and TCP/IP models and highlights how they organize

network functions for efficient data flow.

- [OSI Model](#)
- [TCP/IP Model](#)

Physical Layer

Network communication rests on these physical foundations. It explains network topologies, transmission media, and data transmission modes.

- [Physical Layer](#)
- [Types of Network Topology](#)
- [Modes of Transmission](#)
- [Transmission Media](#)

Data Link Layer

Data link layer is all about how data is reliably transferred between directly connected devices and handles data framing, error detection and correction, flow control, and switching techniques.

- [Data Link Layer](#)
- [Switching Techniques](#)
- [Virtual LAN](#)
- [Link Aggregation](#)
- [Framing](#)
- [Error Detection](#)
- [Error Correction](#)
- [Error Control](#)
- [Flow Control](#)
- [Stop and wait ARQ](#)
- [Sliding Window Protocol- Go Back N](#)
- [Sliding Window Protocol- Selective Repeat](#)
- [Piggybacking](#)

Network Layer

Logical addressing and routing of data packets are explained, including IP addressing, IPv4 vs IPv6, and public and private addresses.

- [Network Layer](#)
- [Classful Network Addressing](#)
- [Classless Network Addressing](#)
- [IPv4 Header Format](#)
- [IPv4 vs IPv6](#)
- [Private vs Public IP addresses](#)

Subnetting

Covers IP address management using subnetting and VLSM to divide large networks into smaller, manageable subnets and calculate address ranges.

- [Subnetting basics](#)
- [Subnet masks](#)
- [Calculate network, broadcast and host addresses](#)
- [Variable Length Subnet Masking \(VLSM\)](#)
- [Advanced VLSM](#)
- [Supernetting](#)

Routing

This subsection explains how data is routed across networks, covering static vs. dynamic routing, routing algorithms, and Network Address Translation (NAT).

- [What is Routing?](#)
- [Static vs Dynamic Routing](#)
- [Link State Routing](#)
- [Distance Vector Routing](#)
- [Network Address Translation\(NAT\)](#)

Network Layer Protocols

Covers key Network Layer protocols such as ARP for device discovery, ICMP for error reporting, DHCP for automatic IP assignment, and dynamic routing protocols like RIP, OSPF, and BGP.

- [Address Resolution Protocol\(ARP\)](#)
- [Reverse Address Resolution Protocol\(RARP\)](#)
- [Dynamic Host Configuration Protocol\(DHCP\)](#)
- [Internet Control Message Protocol\(ICMP\)](#)
- [Internet Group Management Protocol\(IGMP\)](#)
- [Routing Information Protocol\(RIP\)](#)
- [Open Shortest Path First\(OSPF\)](#)
- [Intermediate System to Intermediate System\(IS-IS\)](#)
- [Enhanced Interior Gateway Routing Protocol\(EIGRP\)](#)
- [Border Gateway Protocol\(BGP\)](#)
- [Multiprotocol Label Switching\(MPLS\)](#)
- [Internet Protocol\(IP\)](#)
- [Generic Routing Encapsulation\(GRE\)](#)

Transport Layer

Explains how applications communicate reliably using TCP and UDP, including connection setup, flow control, error checking, and congestion management.

- [Transport Layer](#)
- [TCP Protocol](#)

- [TCP 3-Way Handshake Process](#)
- [Services and Segment structure in TCP](#)
- [TCP Connection Establishment](#)
- [TCP Termination](#)
- [Congestion Control in TCP](#)
- [UDP Protocol](#)
- [TCP vs UDP Protocol](#)
- [Stream Control Transmission Protocol\(SCTP\).](#)
- [Datagram Congestion Control Protocol\(DCCP\).](#)
- [Reliable User Datagram Protocol\(RUDP\).](#)
- [Quick UDP Internet Connections\(QUIC\).](#)

Session Layer

Session management and communication control are discussed through the Session Layer, including protocols such as RPC and PPTP.

- [Session Layer](#)
- [Functions of Session Layer](#)
- [Remote Procedure Call \(RPC\).](#)
- [Point-to-Point Tunneling Protocol](#)

Presentation Layer

Data formatting and security functions are provided by the Presentation Layer, along with protocols such as AFP, NCP, SSL, and MIME.

- [Presentation Layer](#)
- [Services Provided by Presentation Layer](#)
- [Application Filing Protocol\(AFP\).](#)
- [NetWare Core Protocol \(NCP\).](#)
- [Secure Socket Layer \(SSL\).](#)
- [Remote Procedure Call\(RPC\).](#)
- [Point-to-Point Tunneling Protocol\(PPTP\).](#)
- [Multipurpose Internet Mail Extensions\(MIME\).](#)

Application Layer

The Application Layer interfaces with end-users, covering client-server models and core internet services like the web and email.

- [Application Layer](#)
- [Client-Server Model](#)
- [World Wide Web \(WWW\).](#)
- [E-Mail](#)
- [Content Distribution Network\(CDN\).](#)

Application Layer Protocols

Key application protocols such as DNS, HTTP/HTTPS, SMTP, POP3, IMAP, and FTP are discussed in the context of everyday network applications.

- [Domain Name System \(DNS\)](#).
- [File Transfer Protocol \(FTP\)](#).
- [Simple Mail Transfer Protocol \(SMTP\)](#).
- [Simple Network Management Protocol \(SNMP\)](#).
- [HyperText Transfer Protocol \(HTTP\)](#).
- [HyperText Transfer Protocol Secure \(HTTPS\)](#).
- [Post Office Protocol v3 \(POP3\)](#).
- [Internet Message Access Protocol \(IMAP\)](#).
- [Lightweight Directory Access Protocol \(LDAP\)](#).
- [Network Time Protocol \(NTP\)](#).
- [Trivial File Transfer Protocol \(TFTP\)](#).
- [Network News Transfer Protocol \(NNTP\)](#).
- [Message Queuing Telemetry Transport \(MQTT\)](#).
- [Session Initiation Protocol \(SIP\)](#).
- [Server Message Block \(SMB\)](#).

Network Performance and Quality

Network performance and quality are discussed, including congestion control, QoS for multimedia, and traffic-shaping techniques such as token and leaky bucket.

- [Congestion Control](#)
- [QoS and Multimedia](#)
- [Techniques for achieving QoS](#)
- [Token Bucket](#)
- [Leaky Bucket](#)

Network Security

Essential network security concepts such as authentication, encryption, firewalls, IDS/IPS, and VPN are discussed.

- [Authentication](#)
- [Encryption](#)
- [Firewalls](#)
- [MAC Address Filtering](#)
- [Intrusion Detection System](#)
- [Intrusion Prevention System](#)
- [VPN](#)

Wireless and Special Networks

Wireless communication includes technologies like Wi-Fi, Bluetooth, Zigbee, and mobile cellular networks.

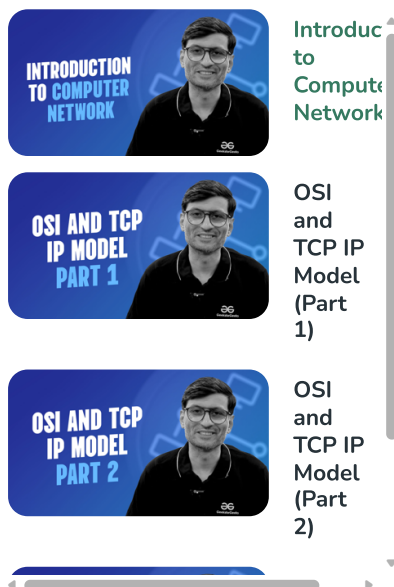
- [Wi-Fi Standards](#)
- [Wireless LAN](#)
- [Bluetooth](#)
- [Zigbee](#)
- [Mobile Networks](#)

Modern Networking Concepts

Cloud networking concepts are discussed, including service models, VPN-based secure access, NFV, and emerging technologies such as 5G and network slicing.

- [Software Defined Networking \(SDN\)](#)
- [Network Function Virtualization \(NFV\)](#)
- [Types of Cloud Services](#)
- [IoT Networking](#)
- [Network Slicing in 5G](#)

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